CORE – 7: Discrete Mathematical Structures – Unit-1

Sets: A set is defined as a collection of **distinct** objects of the same type or class of objects. An object can be numbers, alphabets, names, etc.

Examples of sets are “A set of rivers of India”, “A set of vowels” etc.

We broadly denote a set by the capital letter A, B, C, etc. while the elements of the set by small letter a, b, x, y, etc. If A is a set, and x is one of the elements of A, then we denote it as $x \in A$. Here the symbol $\in$ means -"Element of."

Sets Representation: Sets are represented in two forms:-

(a) Roster or tabular form: In this form of representation we list all the elements of the set within braces { } and separate them by commas.

Example: If A= set of all odd numbers less then 10 then in the roster from it can be expressed as A = {1,3,5,7,9}.

(b) Set Builder form: In this form of representation we list the properties fulfilled by all the elements of the set. We note as {x: x satisfies properties P} and read as 'the set of those entire x such that each x has properties P.'

Example: If B= {2, 4, 8, 16, 32}, then the set builder representation will be:
$B = \{x: x = 2^n, \text{ where } n \in N \text{ and } 1 \leq n \leq 5\}$

Standard Notations:

$x \in A$	x belongs to A or x is an element of set A.
$x \notin A$	x does not belong to set A.
$\emptyset$	Empty Set.
U	Universal Set.
N	The set of all natural numbers.
Z	The set of all integers.
Q	The sets of all rational numbers.
R	The set of all real numbers.
C	The set of all complex numbers.

Cardinality of a Sets: The total number of unique elements in a set is called the cardinality of the set.

Examples:

1. Let P = {k, l, m, n} The cardinality of the set P is 4.
2. Let A is the set of all non-negative even integers, i.e. A = {0, 2, 4, 6, 8, 10.....}.

   Cardinality of A is infinite.

Types of Sets: Sets can be classified into many categories. Some of which are finite, infinite, subset, universal, proper, power, singleton set, etc.

1. Finite Sets: A set is said to be finite if it contains exactly n distinct element where n is a non-negative integer. Here, n is said to be "cardinality of sets." The cardinality of sets is denoted by |A|, # A, card (A) or n (A).

Example:

- Cardinality of empty set ∅ is 0 and is denoted by |∅| = 0
- Set of even positive integers is not a finite set.

2. Infinite Sets: A set which is not finite is called as Infinite Sets.

3. Subsets: If every element in a set A is also an element of a set B, then A is called a subset of B. It can be denoted as $A \subseteq B$. Here B is called Superset of A.

Example: If A= {1, 2} and B= {4, 2, 1} the A is the subset of B or $A \subseteq B$.

Properties of Subsets:

1. Every set is a subset of itself.
2. The Null Set i.e.∅ is a subset of every set.
3. If A is a subset of B and B is a subset of C, then A will be the subset of C. If $A \subset B$ and $B \subset C \Rightarrow A \subset C$
4. A finite set having n elements has 2^n subsets.

4. Proper Subset: If A is a subset of B and $A \neq B$ then A is said to be a proper subset of B. If A is a proper subset of B then B is not a subset of A, i.e., there is at least one element in B which is not in A.

Example:
(i) If A = {2, 3, 4} and B = {2, 3, 4, 5} Then, A is a proper subset of B.
(ii) The null ∅ is a proper subset of every set.

5. Improper Subset: If A is a subset of B and A = B, then A is said to be an improper subset of B.

Example
(i) if A = {2, 3, 4} and B = {2, 3, 4} then, A is an improper subset of B.
(ii) Every set is an improper subset of itself.

6. Universal Set: If all the sets under investigations are subsets of a fixed set U, then the set U is called Universal Set.

Example: In the human population studies the universal set consists of all the people in the world.

7. Null Set or Empty Set: A set having no elements is called a Null set or void set. It is denoted by ∅.

8. Singleton Set: It contains only one element. It is denoted by {s}.

Example: S= {x|x∈N, 7<x<9} = {8}

9. Equal Sets: Two sets A and B are said to be equal and written as A = B if both have the same elements. Therefore, every element which belongs to A is also an element of the set B and every element which belongs to the set B is also an element of the set A.

$A = B \Leftrightarrow \{x \in A \iff x \in B\}$.

If there is some element in set A that does not belong to set B or vice versa then $A \neq B$, i.e., A is not equal to B.

10. Equivalent Sets: If the cardinalities of two sets are equal, they are called equivalent sets.

Example: If A = {1, 2, 6} and B= {16, 17, 22}, they are equivalent as cardinality of A is equal to the cardinality of B. i.e. |A| = |B| = 3

11. Disjoint Sets: Two sets A and B are said to be disjoint if no element of A is in B and no element of B is in A.

Example: If R = {a, b, c} and S = {k, p, m} then, R and S are disjoint sets.

12. Power Sets: The power of any given set A is the set of all subsets of A and is denoted by **P (A)**. If A has n elements, then **P (A)** has $\mathbf{2^n}$ elements.

Example: If A = {1, 2, 3} then, P (A) = {∅, {1}, {2}, {3}, {1, 2}, {1, 3}, {2, 3}, {1, 2, 3}}.

Partitions of a Set: Let S be a nonempty set. A partition of S is a subdivision of S into nonoverlapping, nonempty subsets. Speceficially, a partition of S is a collection $\{A_i\}$ of nonempty subsets of S such that:

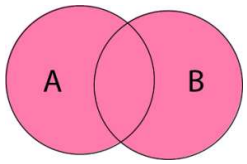
- Each x in S belongs to one of the A_i.
- The sets of $\{A_i\}$ are mutually disjoint; that is, if $A_j \neq A_k$ then $A_j \cap A_k = \emptyset$

Venn Diagrams: Venn diagram is a pictorial representation of sets in which an enclosed area in the plane represents sets.

Operations on Sets: The basic set operations are:

1. Union of Sets: Union of Sets A and B is defined to be the set of all those elements which belong to A or B or both and is denoted by A∪B. i.e. $A \cup B = \{x: x \in A \text{ or } x \in B\}$

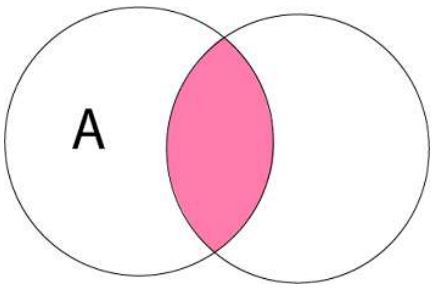
Example: Let A = {1, 2, 3}, B= {3, 4, 5, 6},
then, A∪B = {1, 2, 3, 4, 5, 6}.



2. Intersection of Sets: Intersection of two sets A and B is the set of all those elements which belong to both A and B and is denoted by A ∩ B.

$A \cap B = \{x: x \in A \text{ and } x \in B\}$

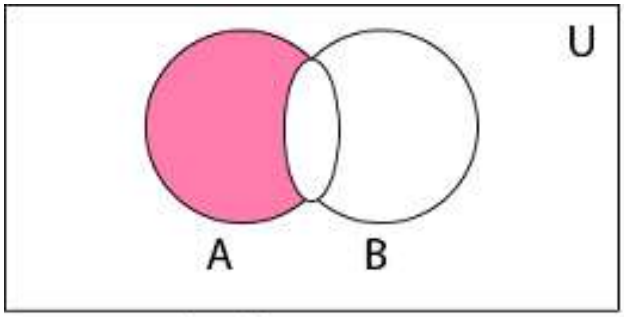
Example: Let A = {11, 12, 13}, B = {13, 14, 15}
A ∩ B = {13}.



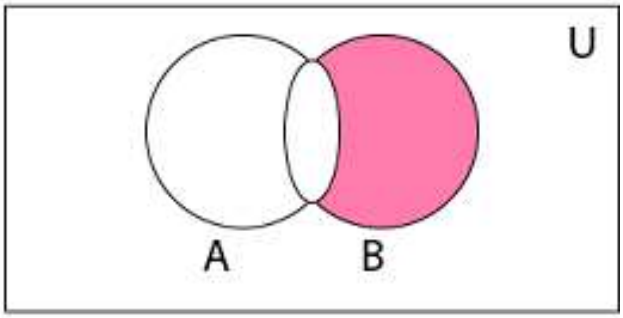
3. Difference of Sets: The difference of two sets A and B is a set of all those elements which belongs to A but do not belong to B and is denoted by A - B.

$A - B = \{x: x \in A \text{ and } x \notin B\}$

Example: Let A= {1, 2, 3, 4} and B = {3, 4, 5, 6} then A - B = {1, 2} and B - A = {5, 6}



A - B

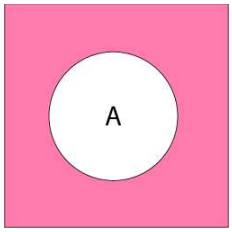


B - A

4. Complement of a Set: The Complement of a Set A is a set of all those elements of the universal set which do not belong to A and is denoted by A^c.

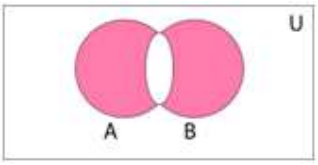
$A^c = U - A = \{x: x \in U \text{ and } x \notin A\} = \{x: x \notin A\}$

Example: Let U is the set of all natural numbers and A = {1, 2, 3}
A^c = {all natural numbers except 1, 2, and 3}.



5. Symmetric Difference of Sets: The symmetric difference of two sets A and B is the set containing all the elements that are in A or B but not in both and is denoted by $A \oplus B$ i.e.

$A \oplus B = (A \cup B) - (A \cap B) = (A - B) \cup (B - A)$



Example: Let A = {a, b, c, d} and B = {a, b, l, m} then,
$A \oplus B$ = {c, d, l, m}

Algebra of Sets: Sets under the operations of union, intersection, and complement satisfy various laws (identities) which are listed in the following table.

Law of Algebra of Sets:

Idempotent Laws	(a) $A \cup A = A$	(b) $A \cap A = A$
Associative Laws	(a) $(A \cup B) \cup C = A \cup (B \cup C)$	(b) $(A \cap B) \cap C = A \cap (B \cap C)$
Commutative Laws	(a) $A \cup B = B \cup A$	(b) $A \cap B = B \cap A$
Distributive Laws	(a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	(b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
De Morgan's Laws	(a) $(A \cup B)' = A' \cap B'$	(b) $(A \cap B)' = A' \cup B'$
Identity Laws	(a) $A \cup \emptyset = A$ (b) $A \cup U = U$	(c) $A \cap U = A$ (d) $A \cap \emptyset = \emptyset$
Complement Laws	(a) $A \cup A' = U$ (b) $A \cap A' = \emptyset$	(c) $U' = \emptyset$ (d) $\emptyset' = U$
Involution Law	(a) $(A')' = A$	

Principle of Inclusions and Exclusions:

Let A, B be any two finite sets. Then $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

Here "include" n (A) and n (B) and we "exclude" $n(A \cap B)$

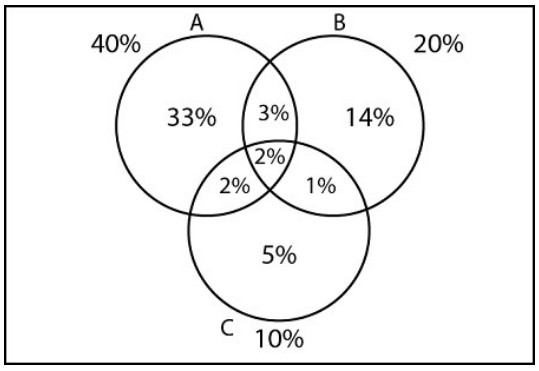
***Example 1**:* Suppose A, B, C are finite sets. Then $A \cup B \cup C$ is finite and

$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$

***Example 2**:* In a town of 10000 families it was found that 40% of families buy newspaper A, 20% family buy newspaper B, 10% family buy newspaper C, 5% family buy newspaper A and B, 3% family buy newspaper B and C and 4% family buy newspaper A and C. If 2% family buy all the newspaper. Find the percentage of of families which buy

1. atleast one newspaper
2. newspaper A only
3. newspaper B only
4. newspaper C only
5. None of A, B, C
6. exactly only one newspaper
7. newspaper A and B only
8. newspaper B and C only
9. newspaper C and A only
10. at least two newspapers
11. at most two newspapers
12. exactly two newspapers

Solution:



1. Number of families which buy atleast one newspaper:
$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$

$n(A \cup B \cup C) = 40+20+10-5-3-4+2 = 60\%$

2. Number of families which buy newspaper A only = 40 - 7 = 33%

3. Number of families which buy newspaper B only = 20 - 6 = 14%

4. Number of families which buy newspaper C only = 10 - 5 = 5%

5. Number of families which buy None of A, B, and C =

$n(A \cup B \cup C)^c = 100 - n(A \cup B \cup C)$
$n(A \cup B \cup C)^c = 100 - 60 = 40\ \%$

6. Number of families which buy exactly only one newspaper = 33 + 14 + 5 = 52%

7. Number of families which buy newspaper A and B only = 3%

8. Number of families which buy newspaper B and C only = 1%

9. Number of families which buy newspaper C and A only = 2%

10. Number of families which buy at least two newspapers = 8%

11. Number of families which buy at most two newspapers = 98%

12. Number of families which buy exactly two newspapers = 6%

Cartesian product of two sets: The Cartesian Product of two sets P and Q is the set of all ordered pairs whose first member belongs to the set P and second member belong to set Q and is denoted by P x Q.

Example: Let P = {a, b, c} and Q = {k, l, m, n}. Determine the Cartesian product of P and Q.

Solution: The Cartesian product of P and Q is

$$P \times Q = \begin{Bmatrix} (a,k), (a,l), (a,m), (a,n) \\ (b,k), (b,l), (b,m), (b,n) \\ (c,k), (c,l), (c,m), (c,n) \end{Bmatrix}$$

Binary Relation: A binary relation R is defined to be a subset of P x Q where P and Q are two non- empty sets. If (a, b) □ R and R □ P x Q then a is related to b by R i.e. aRb. If sets P and Q are equal, then we say R □ P x P is a relation on P e.g.

(i) Let A = {a, b, c} and B = {r, s, t}
Then R = {(a, r), (b, r), (b, t), (c, s)} is a relation from A to B.

(ii) Let A = {1, 2, 3} and B = A
R = {(1, 1), (2, 2), (3, 3)} is a relation on A.

Example1: If a set has n elements, how many relations are there from A to A.
Solution: If a set A has n elements, A x A has n^2 elements. So, there are $2^{n \times n}$ relations from A to A.

Example2: If A has m elements and B has n elements. How many relations are there from A to B and vice versa?

Solution: There are m x n elements; hence there are $2^{m \times n}$ relations from A to B as well as from B to A.

Complement of a Relation: Consider a relation R from a set A to set B. The complement of relation R denoted by R' is a relation from A to B such that

R' = {(a, b): {a, b) ∉ R}.

Example: Consider the relation R from X to Y where X = {1, 2, 3},
Y = {8, 9} and R = {(1, 8) (2, 8) (1, 9) (3, 9)}. Find the complement relation of R.

Solution: X x Y = {(1, 8), (2, 8), (3, 8), (1, 9), (2, 9), (3, 9)}.
So, R' = X x Y – R = {(3, 8), (2, 9)}

Representation of Relations: Relations can be represented in many ways. Some of which are as follows:

1. Relation as a Matrix: Let P = $[a_1,a_2,a_3,.....a_m]$ and Q = $[b_1,b_2,b_3.....b_n]$ are finite sets, containing m and n number of elements respectively. R is a relation from P to Q. The relation R can be represented by m x n matrix M = $[M_{ij}]$, defined as

$M_{ij} = 0$ if $(a_i,b_j) \notin R$
1 if $(a_i,b_i) \in R$

Example: Let P = {1, 2, 3, 4}, Q = {a, b, c, d} and R = {(1, a), (1, b), (1, c), (2, b), (2, c), (2, d)}.

The matrix of relation R is shown as follows.

M_R =

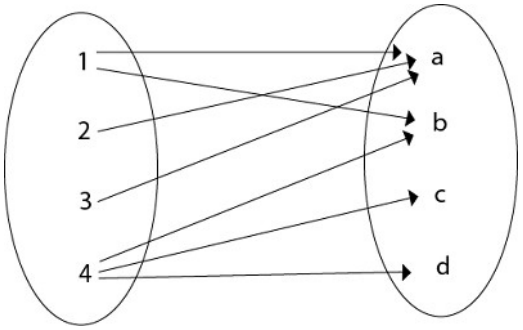
	a	b	c	d
1	1	1	1	0
2	0	1	1	1
3	0	0	0	0
4	0	0	0	0

2. Relation as an Arrow Diagram: If P and Q are finite sets and R is a relation from P to Q. Relation R can be represented as an arrow diagram as follows.

Draw two ellipses for the sets P and Q. Write down the elements of P and elements of Q column-wise in the ellipses. Then draw an arrow from the first ellipse to the second ellipse if a is related to b where a □ P and b □ Q.

Example: Let P = {1, 2, 3, 4} ,
Q = {a, b, c, d}
R = {(1, a), (2, a), (3, a), (1, b), (4, b), (4, c), (4, d)

The arrow diagram of relation R is shown in the fig:



4. Relation as a Table: If P and Q are finite sets and R is a relation from P to Q. Relation R can be represented in tabular form. Make the table which contains rows equivalent to an element of P and columns equivalent to the element of Q. Then place a cross (X) in the boxes which represent relations of elements on set P to set Q.

Example: Let P = {1, 2, 3, 4} ,
Q = {x, y, z, k}
R = {(1, x), (1, y), (2, z), (3, z), (4, k)}.
The tabular form of relation as shown in the fig:

	x	y	z	k
1	x	x		
2			x	
3			x	
4				x

Inverse Relation: Let R be any relation from set A to set B. The inverse of R denoted by R^{-1} is the relations from B to A which consist of those ordered pairs which when reversed belong to R that is:

$R^{-1} = \{(b, a): (a, b) \in R\}$

Example1: A = {1, 2, 3}, B = {x, y, z} and R = {(1, y), (1, z), (3, y). Find R^{-1}.

Solution: $R^{-1} = \{(y, 1), (z, 1), (y, 3)\}$ Clearly $\mathbf{(R^{-1})^{-1} = R}$

Composition of Relations: Let A, B, and C be sets, and let R be a relation from A to B and let S be a relation from B to C. That is, R is a subset of A × B and S is a subset of B × C. Then R and S give rise to a relation from A to C indicated by R◦S and defined by:

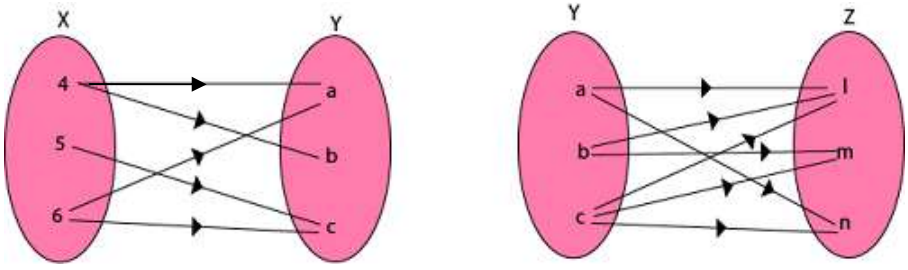
$R \circ S = \{(a, c) \mid$ there exists $b \in B$ **for** which $(a, b) \in R$ and $(b, c) \in S\}$

The relation R◦S is known the composition of R and S; it is sometimes denoted simply by RS.

Let R is a relation on a set A, that is, R is a relation from a set A to itself. Then R◦R, the composition of R with itself. Also, R◦R is sometimes denoted by R^2. Similarly, $R^3 = R^2 \circ R = R \circ R \circ R$, and so on. Thus R^n is defined for all positive n.

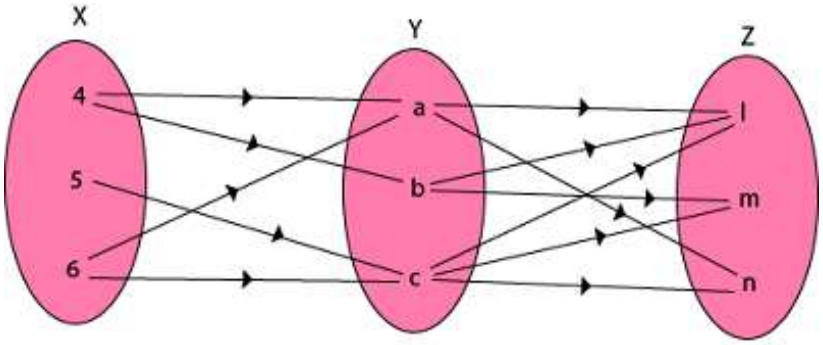
Example1: Let X = {4, 5, 6}, Y = {a, b, c} and Z = {l, m, n}. Consider the relation R_1 from X to Y and R_2 from Y to Z.

$R_1 = \{(4, a), (4, b), (5, c), (6, a), (6, c)\}$
$R_2 = \{(a, l), (a, n), (b, l), (b, m), (c, l), (c, m), (c, n)\}$



Find the composition of relation **(i)** R_1 o R_2 **(ii)** R_1o R_1^{-1}

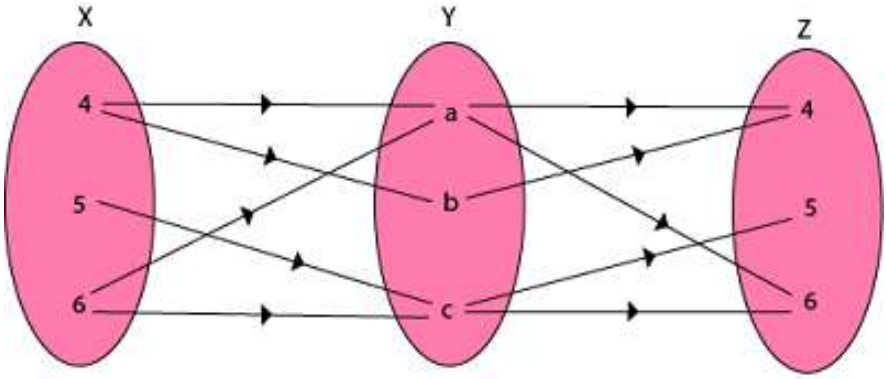
(i) The composition relation R_1 o R_2 as shown in the following fig:



R_1 o R_2 = {(4, l), (4, n), (4, m), (5, l), (5, m), (5, n), (6, l), (6, m), (6, n)}

Fig : R_1 o R_2

(ii) The composition relation $R_1 o R_1^{-1}$ as shown in the following fig:



$R_1 o R_1^{-1}$ = {(4, 4), (4, 6), (5, 5), (5, 6), (6, 4), (6, 6), (6, 5)}

Fig : R_1 o R_1^{-1}

Types of Relations:

1. Reflexive Relation: A relation R on set A is said to be a reflexive if **(a, a)** □ **R** for every **a** □ A.

Example: If A = {1, 2, 3, 4} and relation R (on A) = {(1, 1) (2, 2), (1, 3), (2, 4), (3, 3), (3, 4), (4, 4)}. Is R reflexive?

Solution: The relation is reflexive as for every a □ A. (a, a) □ R,

i.e. (1, 1), (2, 2), (3, 3), (4, 4) □ R.

2. Symmetric Relation: A relation R on set A is said to be symmetric iff

(a, b) □ R => (b, a) □ R.

Example: Let A = {1, 2, 3} and R = {(1, 1), (2, 2), (1, 2), (2, 1), (2, 3), (3, 2)}. Is the relation R symmetric?

Solution: The relation is symmetric as for every (a, b) □ R, we have (b, a) □ R, i.e., (1, 2), (2, 1), (2, 3), (3, 2) □ R but not reflexive because (3, 3) □ R.

3. Transitive Relations: A Relation R on set A is said to be transitive iff

(a, b) □ R and (b, c) □ R => (a, c) □ R.

Example1: Let A = {1, 2, 3} and R = {(1, 2), (2, 1), (1, 1), (2, 2)}. Is the relation transitive?

Solution: The relation R is transitive as for every (a, b) (b, c) belong to R, we have (a, c) ∈ R i.e, (1, 2) (2, 1) ∈ R ⇒ (1, 1) ∈ R.

4. Equivalence Relation: A relation R on a set A is called an equivalence relation if it satisfies following three properties:

1. Relation R is Reflexive, i.e. aRa ∀ a∈ A.
2. Relation R is Symmetric, i.e., aRb => bRa
3. Relation R is transitive, i.e., aRb and bRc => aRc.

Example: Let A = {1, 2, 3, 4} and R = {(1, 1), (1, 3), (2, 2), (2, 4), (3, 1), (3, 3), (4, 2), (4, 4)}. Is R an Equivalence Relation?

Solution:

- Relation R is reflexive as (1, 1), (2, 2), (3, 3) and (4, 4) □ R.
- Relation R is symmetric because whenever (a, b) □ R, (b, a) also belongs to R.
- Relation R is transitive because whenever (a, b) and (b, c) belongs to R, (a, c) also belongs to R.
- So, as R is reflexive, symmetric and transitive, hence, R is an Equivalence Relation.

Antisymmetric Relation: A relation R on a set A is antisymmetric iff (a, b) ∈ R and (b, a) ∈ R then a = b.

Example1: Let A = {1, 2, 3} and R = {(1, 1), (2, 2)}. Is the relation R antisymmetric?

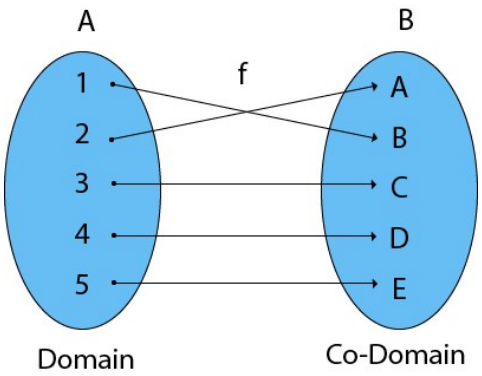
Solution: The relation R is antisymmetric as a = b when (a, b) and (b, a) both belong to R.

Example2: Let A = {4, 5, 6} and R = {(4, 4), (4, 5), (5, 4), (5, 6), (4, 6)}. Is the relation R antisymmetric?

Solution: The relation R is not antisymmetric as 4 ≠ 5 but (4, 5) and (5, 4) both belong to R.

Functions: It is a mapping in which **every element of set A** is uniquely associated with an element of B. The set of A is called Domain of a function and set of B is called Co-domain. In other words,

If P and Q are two non-empty sets, then a function f from P to Q is a subset of P x Q, with two important restrictions;



1. $\forall$ a $\in$ P, (a, b) $\in$ f for some b $\in$ Q
2. If (a, b) $\in$ f and (a, c) $\in$ f then b = c.

Note: There may be some elements of the Q which are not related to any element of set P.

Domain of a Function: Let f be a function from P to Q. The set P is called the domain of the function f.

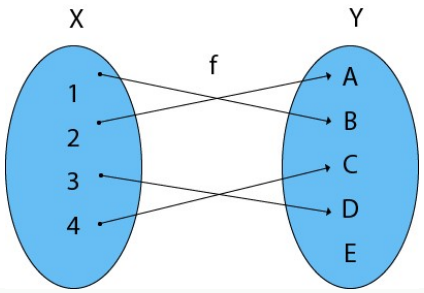
Co-Domain of a Function: Let f be a function from P to Q. The set Q is called Co-domain of the function f.

Range or Image of a Function: The range of a function is the set of picture of its domain. In other words, we can say it is a subset of its co-domain. It is denoted as f (domain).

If f: P → Q, then f (P) = {f(x): x □ P} = {y: y □ Q | □ x □ P, such that f (x) = y}.

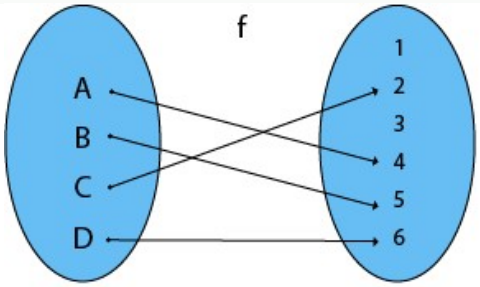
Example: Find the Domain, Co-Domain, and Range of function, where
P = {1, 2, 3, 4} Q = {a, b, c, d, e} f: P → Q = {(1, b), (2, a), (3, d), (4, c)

Solution:

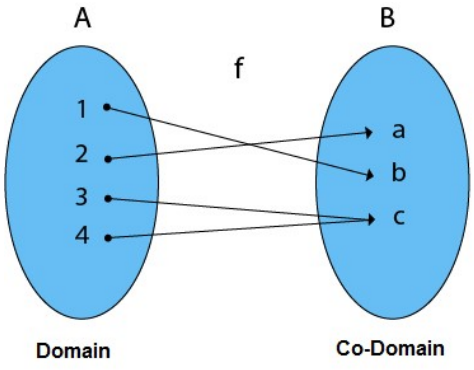


Domain of function: {1, 2, 3, 4}
Co-Domain of function: {a, b, c, d, e}
Range of function: {a, b, c, d}

Types of Functions:

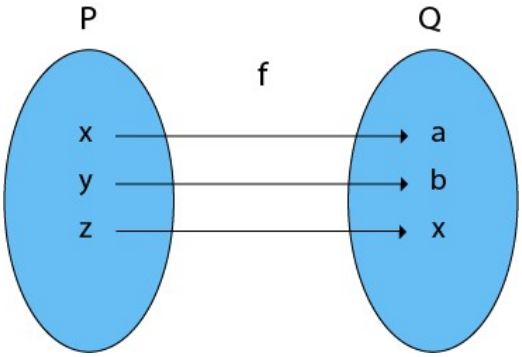


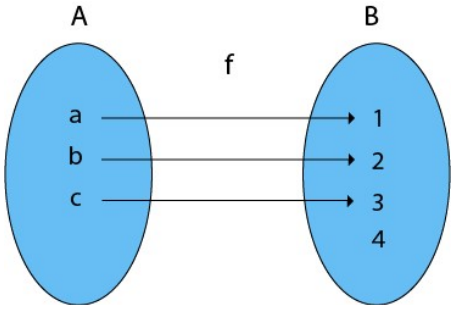
1. Injective (One-to-One) Functions: A function in which one element of Domain Set is connected to one element of Co-Domain Set is called an injective function.



2. Surjective (Onto) Functions: A function in which every element of Co-Domain Set has one pre-image is called an onto function.

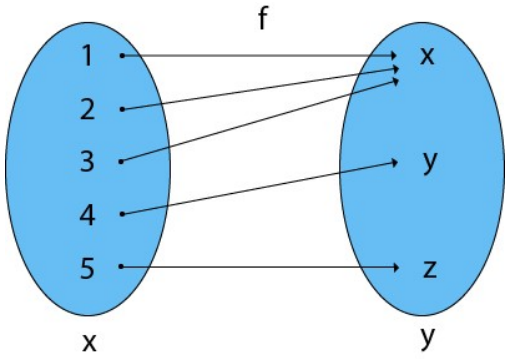
3. Bijective (One-to-One and Onto) Functions: A function which is both injective (one-to-one) and surjective (onto) is called a bijective Function.





4. Into Functions: A function in which there must be an element of co-domain Y that does not have a pre-image in domain X is called an into function.

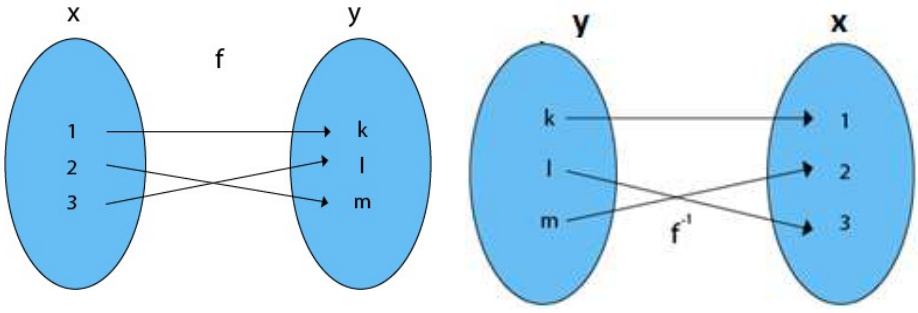
6. Many-One Functions: Let f: X → Y. The function f is said to be many-one functions if there exist two or more than two different elements in X having the same image in Y.



Invertible (Inverse) Functions: A function f: X → Y is invertible if and only if it is a bijective function.

Consider the bijective (one to one onto) function f: X → Y. As f is a one to one, therefore, each element of X corresponds to a distinct element of Y. As f is onto, there is no element of Y which is not the image of any element of X, i.e., range = co-domain Y. The inverse function for f exists if f^{-1} is a function from Y to X.

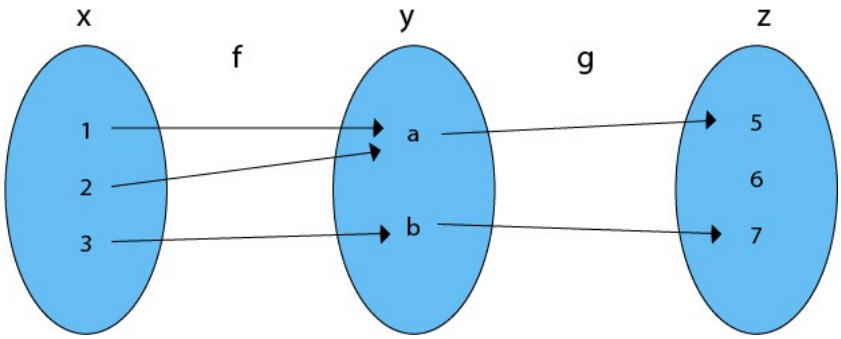
Example:
Consider, X = {1, 2, 3}
Y = {k, l, m} and
f: X→Y such that
f = {(1, k), (2, m), (3, l)



Compositions of Functions: Consider functions, f: A → B and g: B → C. The composition of f with g is a function from A into C defined by (gof) (x) = g [f(x)] and is defined by gof.

To find the composition of f and g, first find the image of x under f and then find the image of f (x) under g.

Example1: Let X = {1, 2, 3}
Y = {a, b}
Z = {5, 6, 7}.
f = {(1, a), (2, a), (3, b)} and
g = {(a, 5), (b, 7)}.
Find the composition gof.



Solution:
(gof) (1) = g [f (1)] = g (a) = 5
(gof) (2) = g [f (2)] = g (a) = 5
(gof) (3) = g [f (3)] = g (b) = 7

Hence, gof = {(1, 5), (2, 5), (3, 7)}

Example2: Consider f, g and h, all functions on the integers, by $f(n) = n^2$, g(n) = n + 1 and h (n) = n - 1. Determine **(i)** hofog **(ii)** gofoh **(iii)** fogoh.

Solution:
(i) hofog (n) = hof (n + 1) = h $[(n+1)^2]$ = $(n+1)^2$ - 1 = n^2 + 1 + 2n - 1 = n^2 + 2n.
(ii) gofoh (n) = gof (n - 1) = g $[(n-1)^2]$ = $(n-1)^2$ + 1 = n^2 + 1 - 2n + 1 = n^2 - 2n + 2.
(iii) fogoh (n) = fog (n - 1) = f [(n - 1) + 1] = f (n) = n^2.

Note:

- If f and g are one-to-one, then the function (gof) (gof) is also one-to-one.
- Composition consistently holds associative property but does not hold commutative property.

Proposition: A Proposition or a statement is a declarative sentence which is either true or false. Propositional logic or Propositional calculus is the branch of logic that deals with propositions (which can be true or false).

Example1: The following statements are all propositions:

- Jawaharlal Nehru is the first prime minister of India.
- If x is an integer, then x^2 is a +ve integer.
- x^2=13

Example2: The following statements are not propositions:

- Please report at 11 a.m. sharp
- What is your name?

Compound Statements: Statements or propositional variables can be combined by means of logical connectives (operators) to form a single statement called compound statements.

Logical connectives:

Symbol	Connective	Name
~	NOT	Negation
∧	AND	Conjunction
∨	OR	Disjunction
⟶	Implies or if...then	Implication or conditional
↔	If and only if	Equivalence or biconditional
↑	NAND	NOT AND
↓	NOR	NOT OR
⊕	X-OR	Exclusive OR

Basic Logical Operations:

1. Negation: It means the opposite of the original statement. If p is a statement, then the negation of p is denoted by ~p and read as 'it is not the case that p.' So, if p is true then ~ p is false and vice versa.
Example: If statement p is Paris is in France, then ~ p is 'Paris is not in France'.

Truth Table	
p	**~ p**
T	F
F	T

2. Conjunction: It means Anding of two statements. If p, q are two statements, then "p and q" is a compound statement, denoted by p ∧ q and referred as the conjunction of p and q. The conjunction of p and q is true only when both p and q are true. Otherwise, it is false.

Truth Table		
p	**q**	**p ∧ q**
T	T	T
T	F	F
F	T	F
F	F	F

3. Disjunction: It means Oring of two statements. If p, q are two statements, then "p or q" is a compound statement, denoted by p ∨ q and referred to as the disjunction of p and q. The disjunction of p and q is true whenever at least one of the two statements is true, and it is false only when both p and q are false.

Truth Table		
p	**q**	**p ∨ q**
T	T	T
T	F	T
F	T	T
F	F	F

4. Implication / if-then (⟶): An implication p⟶q is the proposition "if p, then q." The implication p⟶ q is false only when p is true, and q is false; otherwise, it is always true. In this implication, p is called the hypothesis (or antecedent) and q is called the conclusion (or consequent). **This is equivalent to ~p∨q.**

Truth Table		
p	**q**	**p ⟶ q**
T	T	T
T	F	F
F	T	T

F	F	T

5. If and Only If (↔): If p and q are two statements then "p if and only if q" is a compound statement, denoted as p ↔ q and referred as a biconditional statement or an equivalence. The equivalence p↔q is true only when both p and q are true or when both p and q are false.

Truth Table		
p	**q**	**p ↔ q**
T	T	T
T	F	F
F	T	F
F	F	T

6. NAND (↑): It means negation after ANDing of two statements. Assume p and q be two propositions, Nanding of pand q to be a proposition which is false when both p and q are true, otherwise true. It is denoted by p ↑ q.

Truth Table		
p	**q**	**p ↑ q**
T	T	F
T	F	T
F	T	T
F	F	T

7. NOR (↓): It means negation after ORing of two statements. Assume p and q be two propositions, NORing of p and q is a proposition which is true when both p and q are false, otherwise false. It is denoted by p ↓ q.

Truth Table		
p	**q**	**p ↓ q**
T	T	F
T	F	F
F	T	F
F	F	T

8. XOR (⊕): Assume p and q be two propositions. XORing of p and q is true if p is true or q is true but not both. It is denoted by p ⊕ q.

Truth Table		
p	**q**	**p ⊕ q**
T	T	F
T	F	T
F	T	T
F	F	F

Example1: Prove that X ⊕ Y ≅ (X ∧~Y)∨(~X∧Y). [≅ is read as equivalent to]

Solution: Construct the truth table for both the propositions.

X	**Y**	**X⊕Y**	**~Y**	**~X**	**X ∧~Y**	**~X∧Y**	**(X ∧~Y)∨(~X∧Y)**
T	T	F	F	F	F	F	F
T	F	T	T	F	T	F	T
F	T	T	F	T	F	T	T
F	F	F	T	T	F	F	F

As the truth table for both the proposition is the same, hence

X ⊕ Y ≅ (X ∧~Y)∨(~X∧Y).

Contrapositive: The proposition ~q→~p is called contrapositive of p →q

Converse: The proposition q→p is called the converse of p →q.

Inverse: The proposition $\sim p \rightarrow \sim q$ is called the inverse of $p \rightarrow q$.

Types of propositions based on Truth values: There are three types of propositions when classified according to their truth values

1. **Tautology** – A proposition which is **always true** is called a tautology.
2. **Contradiction** – A proposition which is **always false** is called a contradiction.
3. **Contingency** – A proposition that is **neither a tautology nor a contradiction** is called a contingency.

Example,

1. $p \vee \sim p$ is a tautology.
2. $p \wedge \sim p$ is a contradiction.
3. $p \vee q$ is a contingency.

Predicate Logic: Predicate Logic deals with predicates, which are propositions, consist of variables.

Quantifier: The variable of predicates is quantified by quantifiers. There are two types of quantifier in predicate logic - Existential Quantifier and Universal Quantifier.

Existential Quantifier: If p(x) is a proposition over the universe U, then it is denoted as $\exists x\ p(x)$ and read as "There exists at least one value in the universe of variable x such that p(x) is true. The quantifier $\exists$ is called the existential quantifier. There are several ways to write a proposition, with an existential quantifier, e.g. $(\exists x \in A)p(x)$ or $(\exists x)p(x)$.

Universal Quantifier: If p(x) is a proposition over the universe U, then it is denoted as $\forall x,p(x)$ and read as "For every $x \in U$,p(x) is true." The quantifier $\forall$ is called the Universal Quantifier. There are several ways to write a proposition with a universal quantifier e.g. $\forall x \in A,p(x)$ or $p(x), \forall x \in A$ Or $\forall x,p(x)$.

Negation of Quantified Propositions: When we negate a quantified proposition, i.e., when a universally quantified proposition is negated, we obtain an existentially quantified proposition,and when an existentially quantified proposition is negated, we obtain a universally quantified proposition.

Example: Negate each of the following propositions:

1.$\forall x\ p(x)$	**2. $\exists\ x\ p(x)$**	**3. $\forall x\ p(x) \wedge \exists\ y\ q(y)$**
Sol: $\exists\ x \sim p(x)$	**Sol:** $\forall x \sim p(x)$	**Sol:** $\sim[\forall x\ p(x) \wedge \exists\ y\ q(y)]$ $\cong \sim \forall\ x\ p(x) \vee \sim \exists y q\ (y)$ $(\therefore \sim(p \wedge q) = \sim p \vee \sim q)$ $\cong \exists\ x \sim p(x) \vee \forall y \sim q(y)$

Propositions with Multiple Quantifiers: The proposition having more than one variable can be quantified with multiple quantifiers. The multiple universal quantifiers can be arranged in any order without altering the meaning of the resulting proposition. Also, the multiple existential quantifiers can be arranged in any order without altering the meaning of the proposition.

The proposition which contains both universal and existential quantifiers, the order of those quantifiers can't be exchanged without altering the meaning of the proposition, e.g., the proposition $\exists x \forall y\, p(x,y)$ means "There exists some x such that p (x, y) is true for every y."

Example: Write the negation for each of the following. Determine whether the resulting statement is true or false. Assume U = R (Set of real numbers).

1. $\forall x \exists m(x^2<m)$

Sol: Negation of $\forall x \exists m(x^2<m)$ is $\exists x \forall m (x^2 \geq m)$. The meaning of $\exists x \forall m (x^2 \geq m)$ is that there exists for some x such that $x^2 \geq m$, for every m. The statement is true as there is some greater x such that $x^2 \geq m$, for every m.

2. $\exists m \forall x(x^2<m)$

Sol: Negation of $\exists m \forall x (x^2<m)$ is $\forall m \exists x (x^2 \geq m)$. The meaning of $\forall m \exists x (x^2 \geq m)$ is that for every m, there exists for some x such that $x^2 \geq m$. The statement is true as for every m, there exists some x such that $x^2 \geq m$.

Rules of Inference:

Important Definitions:

1. Argument – A sequence of statements, **premises**, that end with a conclusion.
2. Validity – An argument is **valid** if the conclusion is true whenever all the premises are true.
3. Fallacy – An incorrect reasoning or mistake which leads to invalid arguments.

Structure of an Argument: As defined, an argument is a sequence of statements called premises which end with a conclusion.

If$(p1 \wedge p2 \wedge p3 \wedge \ldots\ldots\ldots \wedge pn) \rightarrow q$ is a tautology, then the argument is termed valid otherwise termed as invalid. The argument is written as –

1st premise
2nd premise
3rd premise
.
.
n^{th} premise
∴ Conclusion

Simple arguments can be used as building blocks to construct more complicated valid arguments. Certain simple arguments that have been established as valid are called Rules of Inference. The most commonly used Rules of Inference are tabulated below –

Rules of Inference	Name	Rules of Inference	Name
P	Addition	$P \wedge Q$	Simplification

$\therefore\ P \vee Q$		$\therefore\ P$	
$\dfrac{P \quad Q}{\therefore\ P \wedge Q}$	Conjuction	$\dfrac{P \to Q \quad Q \to R}{\therefore\ P \to R}$	Hypothetical Syllogism
$\dfrac{P \to Q \quad P}{\therefore\ Q}$	Modus Ponens	$\dfrac{P \vee Q \quad \sim P}{\therefore\ Q}$	Disjunctive Syllogism
$\dfrac{P \to Q \quad \sim Q}{\therefore\ \sim P}$	Modus Tollens	$\dfrac{P \vee Q \quad \sim P \vee R}{\therefore\ Q \vee R}$	Resolution
$\dfrac{P \to Q \quad R \to S \quad P \vee R}{\therefore\ Q \vee S}$	Constructive Dilema	$\dfrac{P \to Q \quad R \to S \quad \sim Q \vee \sim S}{\therefore\ \sim P \vee \sim R}$	Destructive Dilema

Rules of Inference for quantified statements:	
Rule of Inference	Name
$\dfrac{\forall x\ P(x)}{\therefore\ P(c)}$	Universal Instantatiation
$\dfrac{P(c) \text{ for any arbritary } c}{\therefore\ \forall x\ P(x)}$	Universal Generalization
$\dfrac{\exists x\ P(x)}{\therefore\ P(c) \text{ for some } c}$	Existential Instantatiation
$\dfrac{P(c) \text{ for some } c}{\therefore\ \exists x\ P(x)}$	Existential Generalization

Mathematical Induction: Mathematical Induction is a technique which is used to prove a statement, a formula or a theorem is true for every natural number. The technique involves two steps to prove a statement, as stated below −

Step 1(Base step) − It proves that a statement is true for the initial value.

Step 2(Inductive step) − It proves that if the statement is true for the nth iteration (or number *n*), then it is also true for $(n+1)^{th}$ iteration (or number *n+1*).

How to Do It?

Step 1 − Consider an initial value for which the statement is true. It is to be shown that the statement is true for n = initial value.

Step 2 − Assume the statement is true for any value of *n* = *k*. Then prove the statement is true for *n* = *k+1*. We actually break *n* = *k+1* into two parts, one part is *n* = *k* (which is already proved) and try to prove the other part.

Example-1: 3^n - 1 is a multiple of 2 for n = 1, 2, ...

Solution:

Step 1 − For n=1, $3^1-1 = 3-1 = 2$ which is a multiple of 2.
Step 2 − Let us assume 3^n-1 is true for n=k, Hence, 3^k-1 is true

We have to prove that $3^{k+1}-1$ is also a multiple of 2
$3^{k+1}-1 = 3\times3^k-1 = (2+1)\times3^k-1 = 2\times3^k+3^k-1 = (2\times3^k) + (3^k - 1)$
The first part (2×3^k) is certain to be a multiple of 2 and the second part (3^k-1) is also true as our previous assumption.

Hence, $3^{k+1}-1$ is a multiple of 2. So, it is proved that 3^n-1 is a multiple of 2.

Sum of products and product of sums: There are two types of canonical forms:

1. Disjunctive Normal Forms or Sum of Products or (SOP).
2. Conjunctive Normal Forms or Products of Sums or (POS).

Disjunctive Normal Forms or Sum of Products or (SOP): A Boolean expression over ({0, 1}, ∨,∧,') is said to be in disjunctive normal form if it is a join of minterms

Example: $(x_1'\wedge x_2'\wedge x_3') \vee (x_1'\wedge x_2\wedge x_3') \vee (x_1\wedge x_2\wedge x_3)$ is a Boolean expression in disjunctive normal form. Since there are three min-terms $x_1'\wedge x_2'\wedge x_3'$, $x_1'\wedge x_2\wedge x_3$ and $x_1\wedge x_2\wedge x_3$.

Conjunctive Normal Forms or Products of Sums or (POS): A Boolean expression over ({0, 1}, ∨, ∧, ') is said to be in a conjunctive normal form if it is a meet of max-terms

Example: $(x_1\vee x_2\vee x_3)\wedge(x_1\vee x_2\vee x_3)\wedge(x_1\vee x_2\vee x_3)\wedge(x_1'\vee x_2\vee x_3')\wedge(x_1'\wedge x_2'\wedge x_3)$

is a Boolean expression in conjunctive normal form consisting of five max-terms.

Partial ordering: A relation R on set S is called a **partial ordering** or **partial order** if it is **reflexive, anti-symmetric and transitive**. A set S together with a partial order relation R is called a partially ordered set or **POSET** and is denoted as **(S, ≤).** Here the symbol ≤ is used to denote the relation in any poset.

Example:

1. The set N of natural numbers form a POSET under the relation '≤' because firstly $x \leq x$, secondly, if $x \leq y$ and $y \leq x$, then we have $x = y$ and lastly if $x \leq y$ and $y \leq z$, it implies $x \leq z$ for all $x, y, z \in N$.
2. The set N of natural numbers under divisibility i.e., 'x divides y' forms a poset because x/x for every $x \in N$. Also if x/y and y/x, we have $x = y$. Again if x/y, y/z we have x/z, for every $x, y, z \in N$.
3. Consider a set S = {1, 2} and power set of S is P(S). The relation of set inclusion ⊆ is a partial order. Since, for any sets A, B, C in P (S), firstly we have A ⊆ A, secondly, if A ⊆B and B⊆A, then we have A = B. Lastly, if A ⊆B and B ⊆C,then A⊆C. Hence, (P(S), ⊆) is a poset.

Comparable Elements: Two elements a and b of set A are called comparable if either $a \leq b$ or $b \leq a$ holds under relation R.

Non-Comparable Elements: Two elements a and b of set A are called non-comparable if neither $a \leq b$ nor $b \leq a$ holds under relation R.

Example: Consider A = {1, 2, 3, 5, 6, 10, 15, 30} is ordered by divisibility. Determine all the comparable and non-comparable pairs of elements of A.

Solution: The comparable pairs of elements of A are:

{1, 2}, {1, 3}, {1, 5}, {1, 6}, {1, 10}, {1, 15}, {1, 30}
{2, 6}, {2, 10}, {2, 30}
{3, 6}, {3, 15}, {3, 30}
{5, 10}, {5, 15}, {5, 30}
{6, 30}, {10, 30}, {15, 30}

The non-comparable pair of elements of A are:

{2, 3}, {2, 5}, {2, 15}
{3, 5}, {3, 10}, {5, 6}, {6, 10}, {6, 15}, {10, 15}

Linearly Ordered Set or Total order: The set A is called linearly ordered set or totally ordered set, if every pair of elements in A is comparable.

Example: The set of positive integers Z^+ with the usual order ≤ is a linearly ordered set.